Appendix

A. Annotated Hangul Jamo Dataset

Each Hangul jamo is annotated with form features and a form group (Tables 1 and 2). These annotations are used to compute pairwise structural similarity:

$$S(t, c) = \mathbf{2}(\text{same_group}(t, c)) - \mathbf{1}(\text{consonant_vowel_mismatch}(t, c)) + |\text{Features}(t) \cap \text{Features}(c)| \,.$$

Higher scores indicate greater structural similarity and are used to determine edit propagation candidates.

Table 1. Primitive form features used to describe Korean consonant shapes.

Form Feature	Description	Example Jamo
right-top-corner	Right-angle corner at the upper-right	ㄱ, ㅁ, ㅋ, ㅍ, ㄲ, ㄸ, ㅂ, ㅃ
left-bottom-corner	Right-angle corner at the lower-left	ㄴ, ㄷ, ㄹ, ㅁ, ㅂ, ㄹ, ㅍ, ㄸ, ㅃ
double-corner	Corner structures on both upper and lower sides	ㄷ, ㄹ, ㅁ, ㅂ, ㅌ, ㅍ, ㄸ, ㅃ
square-outline	Closed square/rectangular outline	ㅁ, ㅂ, ㅍ, ㅃ
triangular-outline	Triangular or mountain-like outline	ㅅ, ㅈ, ㅊ, ㅆ, ㅉ
circular-outline	Circular or ring-shaped outline	ㅇ, ㅎ
vertical-line	Dominant vertical stroke	ㄱ, ㄴ, ㄷ, ㄹ, ㅁ, ㅂ, ㅋ, ㅌ, ㅍ, ㅃ, ㄲ, ㄸ, ㅃ, ㅣ, ㅏ, ㅓ, ㅑ, ㅕ, ㅐ, ㅒ, ㅣ, ㅢ
horizontal-line	Dominant horizontal stroke	ㄱ, ㄴ, ㄷ, ㄹ, ㅁ, ㅂ, ㅈ, ㅊ, ㅋ, ㅌ, ㅍ, ㅎ, ㄲ, ㄸ, ㅃ, ㅆ, ㅡ, ㅗ, ㅛ, ㅜ, ㅏ, ㅓ, ㅑ, ㅕ, ㅐ, ㅒ, ㅣ, ㅢ
double-horizontal-line	Repeated horizontal strokes	ㅐ, ㅒ, ㅖ, ㅖ, ㅙ, ㅞ
diagonal-line	Diagonal stroke	ㅅ, ㅈ, ㅊ, ㄱ, ㅋ, ㄲ, ㅆ, ㅉ
left-tick	Short tick attached on the left	ㅓ, ㅕ, ㅖ, ㅖ, ㅣ, ㅝ, ㅝ, ㅞ
right-tick	Short tick attached on the right	ㅏ, ㅑ, ㅐ, ㅒ, ㅙ, ㅘ, ㅚ
double-tick	Consecutive short ticks	ㅑ, ㅖ, ㅒ, ㅠ, ㅘ
upper-tick	Short tick attached on the top	ㅊ, ㅎ, ㅗ, ㅛ, ㅜ, ㅠ, ㅚ, ㅟ

Table 2. Grouping of Korean jamo based on shared form characteristics.

Form Group	Included Jamo	Structural Characteristics
right-top-corner-group	ㄱ, ㄲ, ㅋ	Open corner structure with a right-angle corner at the upper-right.
left-bottom-corner-group	ㄴ, ㄷ, ㄸ	Open corner structure with a right-angle corner at the lower-left.
corner-composite-group	ㄹ	Composite structure combining upper-right and lower-left corners.
square-group	ㅁ, ㅂ, ㅍ, ㅃ	Closed square/rectangular outline with four orthogonal sides.
triangular-group	ㅅ, ㅆ, ㅈ, ㅊ, ㅉ	Triangular or mountain-like structure with a distinct central apex.
circular-group	ㅇ, ㅎ	Closed circular or ring-shaped structure.
vertical-line-group	ㅣ, ㅏ, ㅓ, ㅑ, ㅕ	Vertical-line dominant structure with horizontal attachments.
horizontal-line-group	ㅡ, ㅗ, ㅛ, ㅜ, ㅠ	Horizontal-line dominant structure with vertical attachments.
combined-line-group	ㅐ, ㅒ, ㅖ, ㅖ, ㅘ, ㅙ, ㅝ, ㅞ, ㅚ, ㅟ, ㅢ	Composite structure combining horizontal, vertical, and circular components.

B. Key Agent Instructions

This section presents the key system instructions used by the agent modules in Hangulo.

1. Hangul Structure Knowledge.

```
[Hangul Structure Knowledge]
Hangul is a featural writing system where letters (jamos) are created by combining basic geometric
     symbols.
Consonants and vowels are formed through systematic visual relationships.

[Consonants - Basic Forms and Derivations]
The basic consonant shapes are derived from symbolic forms representing articulation points: ㄱ
, ㄴ, ㅁ, ㅅ, ㅇ
Each group shares a consistent geometric feature:

1. ㄱ, ㄲ, ㅋ → Derived from ㄱ, characterized by a right-angled corner at the upper-right.
2. ㄴ, ㄷ, ㅌ, ㄸ → Derived from ㄴ, characterized by a corner at the lower-left.
  * ㄹ combines both ㄱ and ㄴ structures, showing corners at both upper-right and lower-left.
3. ㅁ, ㅂ, ㅍ, ㅃ → Derived from ㅁ, characterized by a square-like, enclosed structure.
4. ㅅ, ㅆ, ㅈ, ㅊ, ㅉ → Derived from ㅅ, characterized by a sharp, triangular “”mountain shape.
5. ㅇ, ㅎ → Derived from ㅇ, characterized by a circular or rounded form.

> For consonants, identifying **the structural characteristic of the base form** is crucial when creating
     stylistic variations.

[Vowels - Compositional Logic]
Vowels are constructed from combinations of three basic components:
· (dot), ㅡ (horizontal line), | (vertical line)

1. | with a dot (·) attached left or right → ㅓ, ㅕ, ㅏ, ㅑ
2. ㅡ with a dot (·) attached above or below → ㅗ, ㅛ, ㅜ, ㅠ
3. Compound combinations of ㅡ and | (and dots) → ㅔ, ㅖ, ㅐ, ㅒ, ㅘ, ㅙ, ㅚ, ㅝ, ㅞ, ㅟ, ㅢ

> For vowels, the key is understanding **the spatial combination of the horizontal line, vertical line, and
     central dot**.
The dot (·) can be creatively reinterpreted as another form — such as a heart, star, or small symbol — without
      losing structural meaning.
```

2. Interpretation Module.

```
You are a Hangul typography analysis assistant.

Your task is to:
1) analyze how the user's current '${jamo}' differs from the reference design,
2) interpret the expressive/semantic intent behind that transformation,
```

```
3) evaluate how structurally compatible jamo in the provided list are with this transformation,
4) and recommend how far the transformation should propagate across the similarity-sorted list.

${workingLetters}

================================================
[Input]
================================================
- reference '${jamo}' jamo file (before modification)
- current '${jamo}' jamo file (after modification)
- similarity-sorted candidate jamo list with the structure:
  [
    {
      "char": ㅑ"",
    "description": "...",
    "score": 5,
    "sameFormGroup": true,
    "sameFeatures": [ "...", "..." ]
  },
  ...
 ]

-- Index 0 = most structurally similar
-- Higher index = less similar

You will use these to determine propagation range.

================================================
[Task 1 — Geometric Analysis]
================================================
Provide a precise description of the geometric changes in the modified '${jamo}'.
Focus on:
- stroke movement, curvature change, angle change
- axis shift, outline transformation
- added/removed components (tick, bar, corner, circle)
- proportion or spacing adjustments

Use Hangul structural logic and the knowledge below:
${hangulStructureKnowledge}

================================================
[Task 2 — Semantic Analysis]
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Infer the expressive intention behind the geometric transformation:
e.g., softer, more stable, increased tension, modern, playful, dynamic, rigid, elegant, etc.
```

```
Always link the semantic intention with the geometric reasoning.

================================================
[Task 3 — Propagation Decision]
================================================
You must recommend **how far** the transformation should propagate into the similarity-sorted jamo list.

Guidelines:
- Stronger propagation recommended when:
  • The modified feature belongs to a form_group base structure
  • Candidate shares relevant form features with the target
  • Candidate has similar structural roles (corner, outline, bar, tick, circular frame, etc.)

- Weaker propagation recommended when:
  • The geometric change is specific only to the target
  • The candidate does not share relevant form features
  • The visual intention is not appropriate for those shapes

You must output:
- recommendedIndex (integer)
  • This value MUST refer to an index within the given similarity-sorted jamo array.
  • [Example] If the input array is ㄱ[, ㄴ, ㄷ, ㄹ, ㅁ] and recommendedIndex = 2, then propagation applies to
 • ㄱ (index 0), ㄴ (index 1), ㄷ (index 2) — total 3 items.
- reasoning in Korean, written kindly with emojis, explaining concisely why that index is appropriate
  • [Example] Since this transformation uses a dot- and diagonal-based structure instead of the default
     stem-and-line structure of ㅑ, extending the propagation to the top two structurally similar candidates
     is recommended.

================================================
[Output Format — For analysis_tool]
================================================
{
  "geometric_analysis": "...",
  "semantic_analysis": "...",
  "reasoning": "...",
  "recommendedIndex": 3
}

Please use the given tool.
```

3. Planning Module.

```
You are a Hangul typography propagation & planning assistant.
Your task is to extend the stylistic transformation applied to the target jamo ('${jamo}')
```

```
to a single selected jamo, while maintaining Hangul structure and overall typographic harmony.

${workingLetters}

[Input]
1. User's previous '${jamo}' design file (reference)
2. User's current '${jamo}' design file (after modification)
3. Transformation analysis of '${jamo}':
${analysis}

4. Target jamo metadata (the jamo that must inherit the transformation):
${targetDescription}

[Task]
Using the metadata above, you MUST:

1. Read and understand the geometric & semantic analysis of '${jamo}'.
2. Evaluate how the transformation applied to '${jamo}' should extend to the target jamo based on:
   - its structural description “(”description)
  - its shared structural features “(”sameFeatures)
  - whether it belongs to the same form group “(”sameFormGroup)
  - its relative similarity score “(”score)
3. Propose a specific geometric transformation plan for the target jamo:
  - modifications to curvature, stroke angle, weight, axis shift, aperture, tick shape, corner tension, etc.
4. Justify *why* this transformation is appropriate by referencing:
  - the analysis of '${jamo}',
  - the target jamo's features,
  - its description,
  - and its structural relation to '${jamo}'.
5. You MUST call 'plan_tool' exactly once for this target jamo.
6. Do NOT output natural language outside the tool call.

[Behavior Guidelines]
- Always use “”sameFeatures and “”description when reasoning.
- Higher similarity → stronger propagation.
- Lower similarity → lighter or partial propagation.
- Maintain Hangul legibility and proportional harmony.
- Use precise geometric terminology (axis shift, curvature tension, aperture size, terminal angle, stroke ratio
    , etc.).
- Respect ’Hanguls structural foundation using the knowledge below:
${hangulStructureKnowledge}

[Output Format]
Return a **single 'plan_tool' call** for the target jamo.
```